Lecture 7: Experiments and what random assignment buys

Empirical Methods for Law and Economics | EDLE

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Today's questions

Did the fine increase late arrivals?

How sure can we be with ten day-care centres?

- By the end of today you will be able to:
  - explain what random assignment guarantees and what it does not;
  - estimate a difference-in-differences from four averages and from a regression;
  - choose standard errors that count the independent units;
  - judge the internal and external validity of an experiment, and design one.

DISCUSS Random assignment

- Gneezy and Rustichini (2000): 10 private day-care centres in Haifa. At the beginning of week 5 a fine was introduced in 6 centres, "selected randomly"; it was removed at the beginning of week 17.

Questions:

1. What does random selection of the six centres guarantee?
2. What does it *not* guarantee, with only ten centres?
3. Why not let the principals decide whether to introduce the fine?

Gneezy, U. and Rustichini, A. (2000), *Journal of Legal Studies* 29(1): 1–17, Section II.B.

Random assignment

- It guarantees that nobody chose the fined centres for a reason related to late arrivals: *on average*, fined and control centres are alike in everything, observed or not.
- With ten centres, "on average" is not "in this draw": the two groups can differ by chance. We check with the weeks before the fine.
- It does not make the 200 weekly counts independent, and it says nothing about other places (external validity).
- Principals who choose the fine may run centres where late arrivals are already rising, or falling: the comparison would mix the fine with the reasons for choosing it.

What we want to know, and what we see

- Each fined centre has two possible numbers of late arrivals: **with** the fine, which we observe, and **without** the fine, which we never observe.
- The effect of the fine is the difference between the two.
- The control centres stand in for the missing number: how fined centres would have behaved without the fine.
- Random assignment makes this stand-in fair on average. The **difference-in-differences** adds a second protection: it removes any constant difference in levels between the two groups.

The data: Table 1 of the paper

- `daycare_fines.dta`: 200 rows, one per centre and week (10 centres $\times$ 20 weeks), transcribed from Table 1.
- Variables: `centre`, `children`, `week`, `late` (late-coming parents in the week), `fine_group` (1 = centres 1–6), `period` (1 = weeks 1–4, 2 = weeks 5–16, 3 = weeks 17–20).
- Notes from the paper:
  - week 11 had four days; the authors multiplied its counts by 5/4 (footnote 6);
  - the study began with 12 centres; two were dropped because their recordings were incomplete (footnote 7).

Gneezy, U. and Rustichini, A. (2000), Table 1, p. 6. Transcribed by the instructor and checked against Table 2.

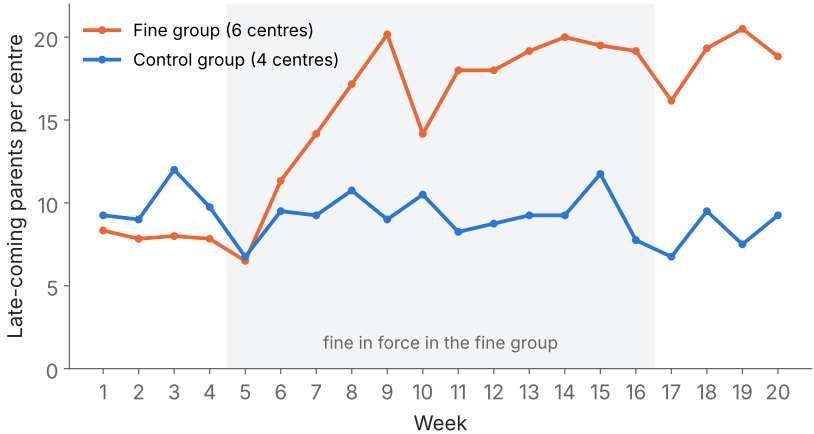
STATA Guided Stata 1: before the fine

```
use "$course_data/daycare_fines.dta", clear
describe
tabstat late children if week <= 4, by(fine_group) statistics(mean n)
preserve
collapse (mean) late, by(fine_group week)
twoway (line late week if fine_group == 1) (line late week if fine_group == 0), ///
    xline(4.5 16.5) legend(order(1 "Fine" 2 "Control"))
restore
```

Weeks 1–4	Fine group	Control group
Late-coming parents per centre and week	8.0	10.0
Children per centre	33.7	33.8
Rows (centre-weeks)	24	16

- `collapse` replaces the data with one row per group and week; `preserve` and `restore` bring the original data back.

Late arrivals, week by week



- Before the fine the two groups move together, with the fine group slightly lower. Then the fine group rises; the control group does not.

PREDICT Four averages

- Weeks 1–4: fine group 8.0, control group 10.0.

Predict, from the figure:

1. the average for each group in weeks 5–16;
2. the difference-in-differences: (fine group, during − before) − (control group, during − before).

- Why subtract the change in the control group instead of comparing the two groups in weeks 5–16 only?

STATA Guided Stata 2: difference-in-differences

```
generate byte during = (period == 2)
tabulate fine_group during if week <= 16, summarize(late) means
regress late i.fine_group##i.during if week <= 16
```

Mean late	Weeks 1–4	Weeks 5–16
Fine	8.00	16.44
Control	10.00	9.23

DiD
$= (16.44 - 8.00) - (9.23 - 10.00) = 9.22$

`regress`, 160 rows	Coef.	(SE)
Constant	10.00	(1.41)
`1.fine_group`	−2.00	(1.82)
`1.during`	−0.77	(1.62)
`1.fine_group#1.during`	9.22	(2.10)

- `##` adds both indicators and their product. The coefficient on the product is the difference-in-differences.

Reading the regression

$$\text{late}_{it} = \beta_0 + \beta_1 \,\text{Fine}_i + \beta_2 \,\text{During}_t + \beta_3 \,(\text{Fine}_i \times \text{During}_t) + u_{it}$$

- $\beta_0 = 10.00$: control group, before.
- $\beta_1 = -2.00$: fine minus control, before. A difference in levels that random assignment did not remove.
- $\beta_2 = -0.77$: change in the control group.
- $\beta_3 = 9.22$: extra change in the fine group, the **difference-in-differences**. About 9 more late-coming parents per centre and week, compared with a starting level of 8.
- Key assumption (**parallel trends**): without the fine, the fined centres would have changed like the control centres. The four weeks before the fine are the evidence we have.

DISCUSS 160 rows, standard error 2.10

Questions:

1. The regression uses 160 rows. How many independent units are there?
2. Which assumption of Lecture 6 is in doubt?
3. What would a referee ask the authors to do?

ANSWERS 160 rows, standard error 2.10

- Ten centres. The fine was assigned to centres, and weeks of the same centre share parents, children and staff.
- Assumption 2, independent observations. The usual standard error treats 160 rows as 160 independent pieces of evidence.
- Count the centres: standard errors **clustered** by centre, or an analysis with one number per centre.

STATA Guided Stata 3: counting centres

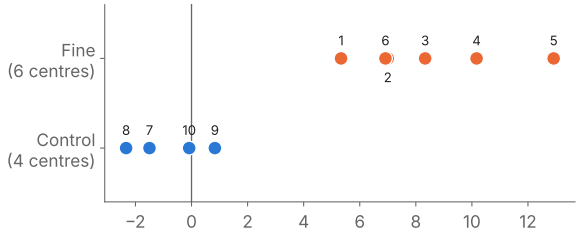
```
regress late i.fine_group##i.during if week <= 16, vce(cluster centre)
preserve
generate late_before = late if week <= 4
generate late_fine = late if week >= 5 & week <= 16
collapse (mean) late_before late_fine, by(centre fine_group)
generate change = late_fine - late_before
list centre fine_group late_before late_fine change
ttest change, by(fine_group)
restore
```

Difference-in-differences: 9.22	SE	t	df	p
Usual SE, 160 rows	2.10	4.39	156	< 0.001
Clustered by centre	1.26	7.30	9	< 0.001
One change per centre, `ttest` (6 vs 4)	1.50	6.15	8	< 0.001

Why the standard error went *down*

- Clustering counts the right units. It does not always make standard errors larger.
- Here the usual formula treats the large differences in levels *between* centres (from 5 to 13 late parents before the fine) as noise. Comparing each centre with itself removes them.
- With 10 clusters the clustered standard error is itself imprecise. The transparent check is the last line: ten numbers, one change per centre, compared across the two groups.
- All three agree: the fine increased late arrivals by about 9 per centre and week. The conclusion does not depend on the choice.

A test that uses only the random assignment



- All six fined centres increased (from $+5.3$ to $+12.9$); the four controls changed between -2.3 and $+0.8$.
- If the fine had no effect, the fined centres would be any six of the ten: 210 equally likely choices. In only one of them do the six largest increases all fall in the fine group: probability $1/210 = 0.005$.
- A **randomization test**: no normal distribution, no large sample; only the fact that the fine was assigned at random.

DISCUSS After the fine was removed

- Weeks 17–20: fine group 18.7, control group 8.3 (before: 8.0 and 10.0). Difference-in-differences: 12.5.

Questions:

1. Late arrivals did not fall back when the fine was removed. Give two explanations.
2. Can these data tell the explanations apart?
3. What does this result imply for a legislator who wants to "try a fine for a while"?

ANSWERS After the fine was removed

- **Information**: the contract was incomplete; the fine told parents that the worst consequence of being late is a small payment, and removing it does not undo that (Section IV.A).
- **Social norms**: the fine turned a favour into a service with a price; "once a commodity, always a commodity" (Section IV.D).
- The data cannot separate them: both predict the same pattern. A design that varies the information given to parents could.
- A sanction can change how people see an obligation. Removing it may not restore the situation before.

DISCUSS Internal and external validity: the referee's view

- Two centres were dropped because their recordings were incomplete (footnote 7).
- Week 11 counts were multiplied by 5/4; there was a one-week holiday after week 14 (footnote 6).
- Teachers recorded late parents; they knew about the fine but not about the study.
- The fine was NIS 10 per child, for 10 minutes or more; a baby-sitter earned NIS 15–20 per hour.
- The fine was announced to parents as a decision of the centre; the announcement does not mention a study (Appendix A).

For each point: does it concern internal validity, external validity, or ethics? Which one would you raise as a main comment?

Gneezy, U. and Rustichini, A. (2000), Section II, footnotes 6–7 and Appendix A.

ANSWERS Internal and external validity: the referee's view

- **Internal**: the two dropped centres matter if they were dropped after assignment and their recording failed because of the fine; a referee would ask in which group they were. The week-11 rescaling and the holiday affect both groups and cancel in the difference-in-differences.

- **Internal**: recording by teachers who knew about the fine could differ between groups; ask how lateness was recorded before the study.

- **External**: a small fine, private centres, one city, January to June 1998. A larger fine might deter; the result shows that a fine *can* backfire, not how often.

- **Ethics**: parents were not asked for consent. Today such a study would need ethical approval and a justification for not informing participants.

DISCUSS Design an experiment

- A court wants to reduce the share of defendants who miss their hearing. Proposal: send text-message reminders a few days before the hearing.

Design the evaluation:

1. What is randomised, and at which level: defendants, days, courtrooms, courts?
2. What is the outcome, and who records it?
3. Two courts or 10,000 defendants: which gives more independent units?
4. What would you check before looking at the results?
5. Could reminders to some defendants affect others?
6. Is it ethical to send reminders to only half of the defendants?

ANSWERS Design an experiment

- Randomise defendants (or hearing dates if messages are sent in batches): many independent units, and the comparison happens within the same courts.
- Outcome: failure to appear, from court records; recorded by staff who do not know who received a message.
- Two courts give two units, however many defendants they have: with randomisation by court, count courts.
- Before the results: balance of characteristics across groups, and the analysis written down in advance.
- Spillovers are small if defendants do not interact; larger with co-defendants or family members.
- Control defendants receive what they receive today. Withholding a reminder whose effect is unknown is defensible; once the effect is known, everyone should get it.

Laboratory

- Worksheet: `do/exercises/07_experiments.do`.
- Data: `daycare_fines.dta`.
- Tasks: the difference-in-differences after the fine was removed; all three periods in one regression; a placebo test on weeks 1–4; late arrivals per 100 children; the randomization test by hand; one written referee comment and one experimental design.
- Bring your draft comments on the paper you are reviewing.

Exit check

1. What does random assignment guarantee, and what does it not guarantee with ten units?
2. Write the difference-in-differences of the fine (9.22) as a sentence.
3. Why can clustered standard errors be smaller than the usual ones?

Next lecture

- Lecture 8 (Friday): legal texts as data. How to turn constitutions into a measure, and how to check that the measure means what we think.
- Read Gentzkow, Kelly and Taddy (2019): Sections 1 and 2; Section 3.5.2.
- Bring your draft comments: feedback and revision in class.